Koopman spectra of trained RNNs: memory, oscillation and steering

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Question

- Can a linear Koopman model of a trained nonlinear RNN show what the network computes?
- Can its eigenvalues and modes steer the network?

Setup

- **Tasks:** 3-bit flip-flop (Sussillo & Barak, 2013), sine generation ($\omega \in [0.1, 0.6]$ rad/step), delayed match-to-sample (DMS).
- **Models:** tanh RNN and GRU, 64 units. 3 seeds per task, 5 for DMS. 20 of 22 models solve their task. The analyses use only these.
- **Dynamics studied:** the autonomous map $F(h) = f(h, x^*)$, with the input held fixed. Data: visited states plus Gaussian noise of size σ, iterated under F.
- **Three linear models of F:**
  - Jacobians at the fixed points (local).
  - Affine DMD on the hidden states (global, observables $\{1, h\}$).
  - Koopman autoencoder, KAE (global, learned observables): MLP encoder and decoder, one dense 16×16 operator K. Loss: reconstruction, latent linearity over 30 steps, state prediction.

Result 1: the global spectrum shows the computation

- **Flip-flop.** Each model has 8 stable fixed points. Their Jacobians are contracting ($|\lambda| \leq 0.77$ GRU, ≤ 0.89 tanh): no local direction stores a bit.
- The KAE has **exactly 3 eigenvalues at 1.000** in 12 of 12 fits, one per bit. DMD has 3–6 eigenvalues above 0.97.
- **Reason.** Each basin of attraction is invariant, so its indicator is a Koopman eigenfunction with eigenvalue 1. Three such functions encode 2^3 basins.
- **Sine.** The KAE has **one pair at $|\lambda| = 1.000$, phase $= \omega$** in 12 of 12 fits. DMD also needs the harmonics 2ω and 3ω.
- **Prediction.** For $\sigma = 0.5$ the KAE predicts the hidden state 1.6–17 times better than DMD at 10 steps, on all 20 models.

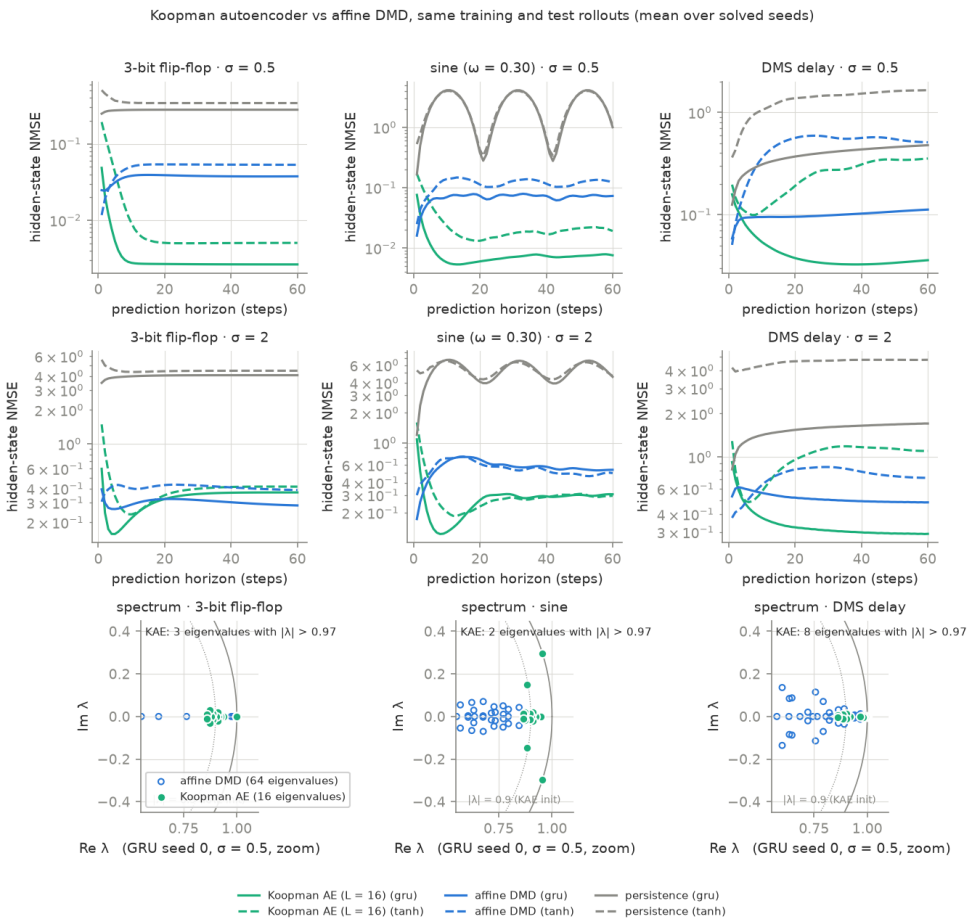


Figure 1: KAE vs affine DMD: prediction error (top) and spectra (bottom).

Result 2: the slow Koopman coordinates steer the network

- **Flip a bit without input.** Move the 3 slow Koopman coordinates a fraction λ towards the target memory pattern, then decode. This is the affine transport of AcT (Rodríguez et al., 2024), applied in Koopman coordinates instead of neurons.
  - Success: **100%** on 6 of 6 models. Other bits changed: **never**.
  - Directions that fail: the gradient of the KAE's slow eigenfunction ($\leq 39\%$, because a basin indicator is flat inside the basin), and the linear response at the nearest fixed point ($\leq 27\%$).
- **Shift the phase.** Rotate the slow pair by α, then decode.
  - Output phase error: median **0.002–0.008 rad**. Amplitude unchanged.
  - A rotation by $k\omega$ equals k steps of the network (error 0.5–3% of the orbit size).
  - The same edit with DMD is 10 times less precise.

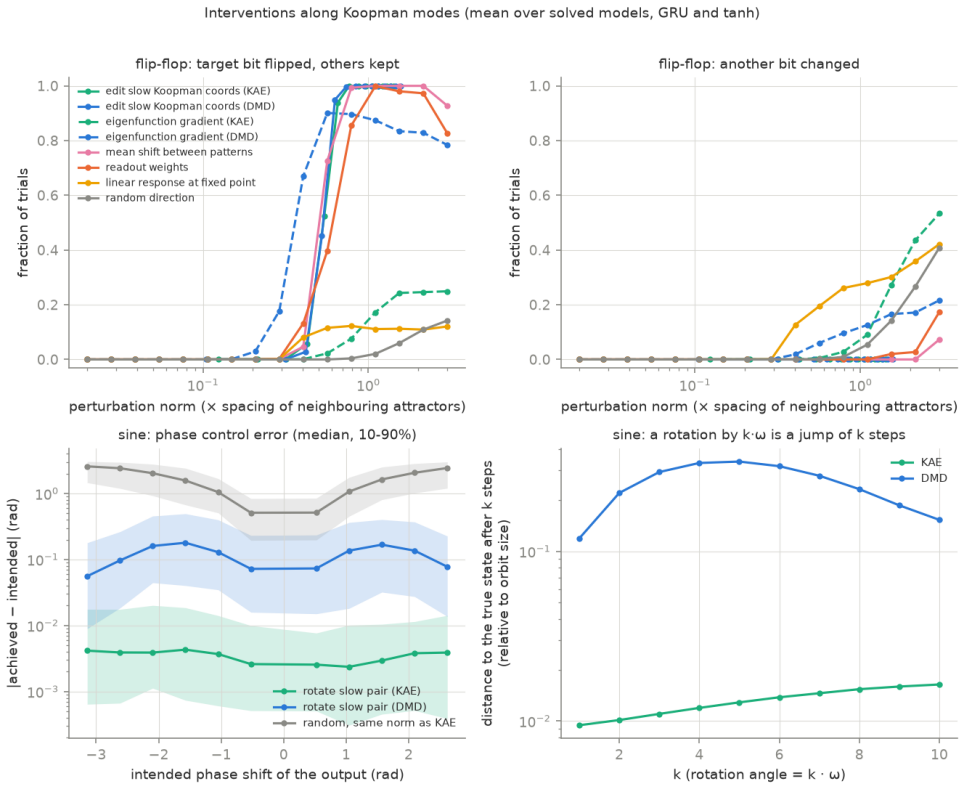


Figure 2: Interventions. Top: flip-flop. Bottom: sine phase control.

Limitations

- The bit edit **does not need less perturbation than a mean shift** between memory patterns. Both use the pattern means, that is, labels.
- The KAE learns only the slow spectrum. Its fast eigenvalues stay near their initialisation (0.9).
- For large perturbations ($\sigma = 2$) the KAE predicts worse than DMD at 60 steps on 4 of 6 flip-flop models.
- Small networks, simple tasks, 3–5 seeds. Hyperparameters were chosen on one model that is also evaluated.

Next steps

- Learn the steering map end to end in Koopman coordinates, with sparsity on the modes, as LinEAS (Rodríguez et al., 2025) does on neurons.
- Apply the analysis to diagonal recurrent cells (ParaGRU, Danieli et al., 2025), whose Jacobians are diagonal.
- Test larger recurrent models, where the answer is not known in advance.

References

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